Numerical Investigation of Flow Past a Circular Cylinder using OpenFOAM

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Declaration of Originality

I hereby declare that this report is my own original work. The underlying research, the design of the parametric meshing and automation framework, and the analysis and conclusions presented herein are my own. All sources consulted during this investigation have been properly cited in accordance with academic conventions, and no portion of this work has been submitted previously for assessment at this or any other institution.

In the preparation of this report, an AI-based language model (Anthropic's Claude) was used as a writing and coding assistance tool, specifically to help structure and phrase sections of the written report, and to support the optimisation, debugging, and documentation of the supporting OpenFOAM configuration files and shell/Python scripts developed for this project.

Signature: _____ Date: September 30, 2026

Abstract

Computational studies of flow past a circular cylinder are frequently limited, in practice, not by solver capability but by the effort required to generate a consistent, high-quality mesh across the range of Reynolds numbers and resolutions needed for a credible parametric campaign. This project addresses that limitation by developing a fully parameterised, block-structured meshing and case-automation framework for incompressible flow past a circular cylinder in OpenFOAM, in which every geometric and grading quantity in the mesh is derived analytically, via OpenFOAM's `#calc` macro, from a small set of independent control parameters.

The mesh itself adopts a nested double O-grid topology: an inner ring of hexahedral blocks dedicated to resolving the near-wall viscous region, and an outer ring that buffers the transition into a far-field H-grid, with the far-field cell counts and gradings derived from a matched interface cell width so that cell size remains continuous across every internal block boundary. This mesh-generation logic is exercised at scale by a dedicated driver script that instantiates, configures, meshes, and solves an arbitrary number of cases from a single tabulated specification, applying a template-copy instantiation pattern and per-case, per-stage fault isolation suited to unattended execution of large parametric sweeps.

Building on this infrastructure, the project further specifies a verification and validation protocol, covering mesh independence, temporal convergence, domain-size sensitivity, and validation against published benchmark data, and shows that each of these studies reduces, within the present framework, to varying one or two parameters of the case specification rather than requiring custom manual intervention.

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1
Introduction

1.1 Motivation and Background

The study of flow around bluff bodies represents a foundational pillar of applied fluid mechanics. Unlike streamlined bodies, bluff geometries generate separated wake regions whose unsteady dynamics strongly influence the forces experienced by structures across a vast range of engineering applications. Among the many canonical bluff-body configurations, the circular cylinder has attracted particular and sustained scientific interest for more than a century, primarily because of the exceptional clarity with which it exhibits the essential phenomena of boundary-layer separation, periodic vortex formation, and pressure-induced drag.

In practical engineering, cylindrical or near-cylindrical cross-sections appear in bridge piers, offshore risers, heat-exchanger tube bundles, overhead transmission lines, chimney stacks, and marine mooring cables. In each of these contexts, the alternating pressure loading associated with periodic wake shedding can drive resonant structural vibrations whose amplitude grows dramatically when the shedding frequency couples with a natural frequency of the structure—a mechanism known as lock-in or vortex-induced vibration (VIV). The consequences range from fatigue damage and reduced service life to catastrophic structural failure, as evidenced by the 1940 collapse of the Tacoma Narrows Bridge [6]. Predictive computational models that reliably capture shedding frequency, mean drag, and fluctuating lift are therefore indispensable design tools.

1.2 Research Objectives and Report Structure

The overarching objective of this investigation is to develop a progressively refined, multi-fidelity computational framework for the analysis of incompressible flow past a circular cylinder. Central to this objective is the creation of a fully automated, heavily parameterized OpenFOAM case structure. By driving the domain geometry, mesh topology, and solver execution entirely through an automated pipeline, this study aims to systematically escalate modelling complexity while maintaining mathematical consistency across all fidelity levels. This approach bridges the gap between traditional computational fluid dynamics (CFD) and data-driven surrogate modelling.

To realise this overarching goal, the specific objectives are defined as follows:

1. **Development of a Parametric Meshing Architecture:** Construct a robust, automated meshing utility using a bespoke `geometryProperties` dictionary. This objective involves formulating mathematically linked relationships that control near- wall boundary layer clustering, double O-grid transition smoothing, and far- field spatial discretization, allowing rapid iteration across diverse flow regimes without manual grid regeneration.

2. **Baseline Laminar Configuration and Validation:** Establish a rigorously verified laminar solver configuration within the automated environment. This foundational step will validate mean and unsteady aerodynamic force coefficients, Strouhal numbers, and wake topologies against established theoretical and experimental benchmarks from the open literature.

3. **Development of a 3D Large Eddy Simulation (LES) Framework:** Formulate and validate an unsteady LES framework on three-dimensional computational domains. The automated script will be utilized to extrude the domain spanwise and heavily refine the wake region, focusing on the sensitivity of predicted flow statistics and coherent structures to the choice of subgrid-scale (SGS) models.

Structure of the Report

The remainder of this report is organised as follows to systematically address the aforementioned objectives.

Chapter 2 presents a focused review of the pertinent literature, establishing the current state-of-the-art in cylinder wake dynamics, multi-fidelity CFD, and machine learning applications in fluid mechanics.

Chapter 3 outlines the computational methodology, providing an in-depth discussion of the domain geometry, meshing strategy, boundary conditions, and the spatial and temporal numerical discretization schemes employed in OpenFOAM.

Chapter 4 describes the campaign-automation framework built around this meshing strategy, detailing the directory architecture, the tabulated case-specification schema, the case-instantiation and parameter-injection pipeline, and the fault-tolerant execution and logging procedures used to generate and run an arbitrary number of parametrically distinct cases without manual intervention.

Chapter **??** sets out the verification and validation methodology by which the accuracy and reliability of the resulting simulations are to be established, outlining how mesh-independence, temporal-convergence, and Reynolds-number sweep studies are conducted within the existing framework by varying a small number of case parameters.

Finally, Chapter 6 synthesises the key findings and contributions of the work, discusses the limitations of the present framework, and identifies directions for future development and extension of the methodology.

2

Background

This chapter presents the foundational concepts necessary for understanding the scope and significance of this research. It introduces the operational principles of fixed-bed reactors, their applications in industry, underlying transport phenomena, and key computational tools relevant to their modeling.

2.1 Classical Experimental Studies

The scientific exploration of circular-cylinder wakes can be traced to the systematic flow visualisation experiments of von Kármán and Rubach in the early twentieth century, who provided the first theoretical account of the stability characteristics of the staggered double vortex row now bearing von Kármán's name [20]. Roshko's landmark experimental work [38] established the empirical relationship between Strouhal number and Reynolds number across a wide parameter range and introduced the concept of a universal wake Strouhal number based on wake width rather than cylinder diameter.

Williamson's contributions [47, 49] refined understanding of the transitions between distinct wake regimes, documenting the discontinuous jumps in Strouhal number associated with changes in vortex-filament topology (the so-called A- and B-mode three-dimensional instabilities), and providing meticulously measured force coefficient and frequency data that have since become the primary validation reference for computational studies. Zdravkovich's two-volume compilation [51] surveys the full body of experimental knowledge on cylinder aerodynamics and remains an indispensable reference.

Achenbach [1] investigated the high-Reynolds-number regime in detail, documenting the drag crisis and its sensitivity to surface roughness and free-stream turbulence intensity. More recently, quantitative techniques such as particle-image velocimetry (PIV) have been applied to cylinder wakes to provide spatially resolved velocity and vorticity fields suitable for direct comparison with simulation outputs [32].

2.2 Reynolds Number Regime Classification

The recognised classification of cylinder wake regimes provides the organisational framework for the present investigation. Following [51] and [49], the principal regimes

may be summarised as follows:

- **Creeping / Stokes flow** ($Re < 1$): Flow is fully attached and reversible; viscous forces dominate.
- **Steady separated wake** ($1 < Re < 47$): A recirculating bubble forms; the flow remains steady and symmetric.
- **Laminar periodic shedding** ($47 < Re < 190$): Periodic two-dimensional shedding; well-defined Strouhal number.
- **Three-dimensional transition** ($190 < Re < 1000$): Spanwise instabilities (modes A and B) introduce three-dimensional structure.
- **Irregular turbulent wake** ($Re > 10^3$): Increasingly disordered and broadband shedding.
- **Drag crisis** ($Re \approx 3 \times 10^5$–5×10^5): Boundary-layer transition, delayed separation, and sharp drag reduction.

2.3 Computational Studies of Laminar Cylinder Flow

Two-dimensional laminar simulations of cylinder flow have been reported by numerous investigators. Posdziech and Grundmann [34] employed a spectral- element method on extremely large computational domains to provide resolution-independent reference values of $\overline{C}_D$, $C_{L,\mathrm{rms}}$, and St for $Re = 20$–250, accounting carefully for blockage effects. Their data have been widely adopted as reference values for code-verification exercises. Barkley and Henderson [3] used Floquet stability analysis to characterise the three-dimensional wake instabilities, establishing accurate threshold Reynolds numbers for mode A and mode B onset. Liu et al. [25] performed detailed time-dependent simulations confirming the accuracy of the von Kármán shedding mechanism and reporting pressure distributions that compare favourably with experiment.

Finite-volume-based studies using OpenFOAM or comparable tools have reproduced these benchmark results with second-order schemes on structured and block-structured meshes [37, 12], confirming that the platform is capable of accurate laminar predictions when mesh resolution and domain extent are chosen appropriately.

2.4 RANS Modelling of Turbulent Cylinder Flow

RANS simulations of cylinder flow in the turbulent regime present well-known difficulties. Franke and Rodi [17] demonstrated that standard linear eddy-viscosity models tend to overpredict eddy viscosity in the separating shear layers, suppress the shedding oscillation, and misrepresent the recirculation length. The two-layer models of that era showed modest improvements near the surface but poor wake recovery. Catalano et al. [10] compared several RANS models with LES reference data at $Re = 10^6$ and found substantial model-to-model scatter, with all RANS closures underestimating base pressure suction and over-predicting recirculation length to varying degrees.

The k–ω SST model of [27] has shown superior performance in separated and adverse-pressure-gradient flows compared with k–ε variants, owing to its selective use of the ω-equation in near-wall regions and the ε-formulation in the far field. Nevertheless, even this model encounters difficulties with the highly curved streamlines in the near wake of a cylinder.

2.5 Large Eddy Simulation of Cylinder Wakes

LES approaches have demonstrated considerably higher predictive fidelity for cylinder wakes. The pioneering LES study of [4] established that LES with the Smagorinsky subgrid model could reproduce experimental velocity spectra and force statistics at $Re = 3900$ provided the mesh was sufficiently resolved in the azimuthal direction. Subsequent investigations at the same Reynolds number, including those by [16] and [32], have established a detailed dataset against which new subgrid models and numerical schemes can be assessed. Breuer [7, 8] conducted extensive parametric LES studies over a wide Reynolds number range, identifying the sensitivity of predicted recirculation length and base pressure to subgrid model, mesh resolution, and spanwise domain extent.

The dynamic version of the Smagorinsky model, first proposed by [18] and reformulated by [23], removes the necessity of prescribing the Smagorinsky constant by locally and dynamically computing it from the resolved velocity field. The WALE (Wall-Adapting Local Eddy-viscosity) model [28] additionally vanishes naturally at solid walls without explicit damping functions, offering practical advantages for wall-resolved cylinder simulations.

2.6 Data-Driven and AI-Assisted CFD

The integration of machine learning with computational fluid dynamics has accelerated rapidly over the past decade. Supervised neural network architectures have been trained on CFD datasets to construct surrogate models capable of predicting force coefficients and wake statistics at a fraction of the cost of full simulations [9, 22]. Physics-Informed Neural Networks (PINNs), introduced by [35], embed the governing partial differential equations as soft constraints in the loss function, enabling data-sparse reconstruction of flow fields and identification of physical parameters. Their application to cylinder-wake reconstruction from sparse sensor measurements has demonstrated the potential of the approach for real-time flow estimation [19].

Proper Orthogonal Decomposition [26, 42] extracts a low-dimensional basis from an ensemble of flow snapshots, capturing the dominant coherent structures—including the vortex-shedding mode—in the leading modes. The resulting reduced-order model (ROM) can reproduce the unsteady dynamics at orders-of-magnitude lower cost than the full simulation, making it attractive for real-time control and optimisation applications [29, 43]. The multi-fidelity dataset generated by the present investigation provides an ideal resource for exploring these techniques systematically.

2.7 Governing Equations for Incompressible Flow

The motion of an incompressible, Newtonian fluid is governed by the continuity equation expressing conservation of mass,

$$\nabla \cdot \mathbf{u} = 0, \tag{2.1}$$

and the Navier–Stokes momentum equations,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u}, \tag{2.2}$$

where $\mathbf{u}(\mathbf{x}, t)$ is the velocity field, $p(\mathbf{x}, t)$ the kinematic pressure (pressure divided by density), ρ the (constant) fluid density, and $\nu = \mu/\rho$ the kinematic viscosity. In the current study, air at standard conditions is the working fluid, with $\nu = 1.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$. The cylinder diameter is $D = 0.1\,\mathrm{m}$, and the free-stream velocity for a desired Reynolds number follows from re-arranging Reynolds equation,

$$U_\infty = \frac{Re \cdot \nu}{D}. \tag{2.3}$$

2.8 Reynolds Decomposition and the RANS Equations

In turbulent flow, each flow variable ϕ is decomposed into a time-averaged component $\overline{\phi}$ and a fluctuating component ϕ', such that $\phi = \overline{\phi} + \phi'$ and $\overline{\phi'} = 0$. Applying this Reynolds decomposition to Equations (2.1) and (2.2) and time-averaging yields the Reynolds-Averaged Navier–Stokes (RANS) equations:

$$\nabla \cdot \overline{\mathbf{u}} = 0, \tag{2.4}$$

$$\frac{\partial \overline{\mathbf{u}}}{\partial t} + (\overline{\mathbf{u}} \cdot \nabla)\overline{\mathbf{u}} = -\nabla \overline{p} + \nu \nabla^2 \overline{\mathbf{u}} - \nabla \cdot \boldsymbol{\tau}^R, \tag{2.5}$$

where the Reynolds stress tensor is defined as

$$\tau_{ij}^R = \overline{u_i' u_j'}. \tag{2.6}$$

The Reynolds stress tensor introduces six additional unknowns into a system of four equations, constituting the well-known *closure problem*. Linear eddy-viscosity models resolve this by invoking the Boussinesq hypothesis,

$$\tau_{ij}^R = -\nu_t \left(\frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial \overline{u}_j}{\partial x_i} \right) + \frac{2}{3} k \delta_{ij}, \tag{2.7}$$

where ν_t is the turbulent (eddy) viscosity and $k = \frac{1}{2}\overline{u_i' u_i'}$ is the turbulent kinetic energy.

2.9 Force Coefficients and Strouhal Number

The macroscopic aerodynamic forces acting on the cylinder surface, which arise from the integration of pressure fields and viscous shear stresses, are conventionally expressed in non-dimensional form. The instantaneous drag and lift coefficients, denoted as C_D and C_L respectively, are defined as:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 D L}, \qquad C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 D L}, \tag{2.8}$$

where F_D and F_L represent the total streamwise (drag) and cross-stream (lift) forces. The denominator term $\frac{1}{2}\rho U_\infty^2$ is the free-stream dynamic pressure, D is the cylinder diameter, and L defines the spanwise length of the computational domain (taken as unity in two-dimensional simulations).

Furthermore, the unsteady, oscillatory dynamics of the cylinder wake, characterized by the alternating separation of von Kármán vortices, are quantified using the dimensionless Strouhal number (St). This parameter relates the dominant vortex shedding frequency to the macroscopic length and velocity scales of the flow:

$$St = \frac{f_s D}{U_\infty}, \tag{2.9}$$

where f_s is the primary shedding frequency in Hertz. In this study, f_s is extracted by performing a Fast Fourier Transform (FFT) on the fluctuating lift coefficient signal over a statistically stationary time window.

3

Computational Methodology

3.1 Overview

The computational methodology adopted in this study is organised around four principal components: (i) definition of the physical geometry and the computational domain, (ii) parametric mesh generation using OpenFOAM's native `blockMesh` utility, (iii) specification of boundary conditions, and (iv) configuration of the numerical schemes and solver settings. Because the circular cylinder is a geometrically simple body, the entire mesh is constructed analytically within a single `blockMeshDict` file without recourse to external CAD software. This approach guarantees full reproducibility and allows all geometric and mesh-density parameters to be varied through a single, centrally managed `geometryProperties` dictionary, whose entries are evaluated at run time via OpenFOAM's `\#calc` macro.

Unlike a single-block or single-O-grid discretisation, the near-cylinder region is resolved using a *nested double O-grid* topology: an inner ring of four hexahedral blocks dedicated to the viscous sublayer, and an outer ring of four hexahedral blocks acting as a graded buffer before the mesh transitions into an eight-block far-field H-grid (Section 3.3). Furthermore, unlike a strictly two-dimensional formulation obtained through an `empty` front/back patch, the present mesh is genuinely three-dimensional: the `frontAndBack` patch is assigned the `wall` type and the spanwise direction is resolved with $N_z = 10$ cells, so that spanwise flow structures can, in principle, be captured rather than suppressed by construction. Case-to-case variation of both the physical parameters (Reynolds number, solver, time-integration scheme) and the mesh-density parameters is orchestrated externally by the `runCases.sh` driver script operating on a tabulated case list, `cases.csv` (Section 3.8), rather than by manual editing of individual case dictionaries.

For more complex bodies, bluff shapes with surface curvature discontinuities, sharp trailing edges, or three-dimensional features, this strategy would be extended by first generating a surface model in a CAD environment and subsequently using `snappyHexMesh` to cut and refine a background mesh around the imported geometry; for the circular cylinder considered here, the fully analytic block-structured approach is preferred for the exactness and reproducibility it affords.

3.2 Physical Geometry and Domain Dimensions

The cylinder diameter D is the single master length scale of the mesh; all other geometric quantities are expressed in `geometryProperties` as exact multiples of D, so that a single change to D propagates automatically and consistently to every block vertex, arc definition, and domain boundary. For the baseline configuration examined in this chapter, $D = 0.1\,\text{m}$ and $R = D/2 = 0.05\,\text{m}$. The domain extents are summarised in Table 3.1.

Table 3.1: Computational domain dimensions, non-dimensionalised by the cylinder diameter D, for the baseline configuration ($D = 0.1\,\text{m}$).

Parameter	Symbol	Expression	Value (m)
Upstream length	X_{min}	$-10D$	-1.000
Downstream length	X_{max}	$+20D$	$+2.000$
Cross-stream lower bound	Y_{min}	$-10D$	-1.000
Cross-stream upper bound	Y_{max}	$+10D$	$+1.000$
Spanwise extent (front)	Z_{min}	$-0.5D$	-0.050
Spanwise extent (back)	Z_{max}	$+0.5D$	$+0.050$

The cross-stream boundaries at $Y_{min} = -10D$ and $Y_{max} = +10D$ yield a domain blockage ratio of

$$\beta = \frac{D}{Y_{max} - Y_{min}} = \frac{D}{20D} = 5.0\,\%. \tag{3.1}$$

This value sits at the upper edge of the threshold ($\beta < 5\,\%$) commonly regarded as sufficient to neglect blockage-correction effects when comparing against unconfined experimental or reference numerical data; it is not comfortably below this threshold, and a domain-sensitivity check (e.g. repeating a baseline case with Y_{max}, Y_{min} scaled outward) is advisable before the blockage-free assumption is relied upon quantitatively in the results chapters.

The independent and derived geometric parameters defined in `geometryProperties` and referenced throughout `blockMeshDict` are summarised in Table 3.2.

Table 3.2: Parametric variables defined in `geometryProperties` and referenced throughout `blockMeshDict` for the baseline configuration.

Variable	Definition	Value	Role
`D`	input	0.1 m	Cylinder diameter
`R`	D/2	0.05 m	Cylinder radius
`R_prism`	D*R_prism_mult	0.06 m	Inner O-grid (prism) boundary radius
`R_sq`	1.5*D	0.15 m	Outer O-grid boundary radius
`X_min`	-10*D	-1.0 m	Inlet face x-coordinate
`X_max`	20*D	$+2.0$ m	Outlet face x-coordinate
`Y_min`	-10*D	-1.0 m	Bottom face y-coordinate
`Y_max`	10*D	$+1.0$ m	Top face y-coordinate
`Z_min`	-0.5*D	-0.05 m	Front face z-coordinate
`Z_max`	0.5*D	$+0.05$ m	Back face z-coordinate
`ds_outer`	3*D/N_c	5.0×10^{-3} m	Matched cell width at O-grid/H-grid interface
`C`	$1/\sqrt{2}$	0.7071	Diagonal projection constant
`R_C`	D*0.5*C	3.536×10^{-2} m	Cylinder-block diagonal vertex offset
`R_prism_C`	D*R_prism_mult*C	4.243×10^{-2} m	Prism-block diagonal vertex offset
`l_Ref`	D	0.1 m	Reference length for Re and St
`A_Ref`	1.0*D*D	1.0×10^{-2} m^2	Reference area for C_D and C_L

The diagonal projection constant $\texttt{C} = 1/\sqrt{2} \approx 0.7071$ gives the Cartesian coordinate (equal in x and y) of the four 45°-diagonal vertices bounding both the cylinder-wall block (`R_C`) and the prism block (`R_prism_C`), ensuring that the circular arc edges can be specified with a single mid-arc point per edge and that the resulting hexahedral faces remain geometrically regular at the block corners.

3.3 Parametric Double O-Grid Topology

In place of a single H-grid block deformed to conform to the cylindrical wall, which produces severely skewed, high-aspect-ratio cells at the four points where a rectangular block boundary is tangent to a circular surface, the near-body mesh is constructed from two concentric rings of four hexahedral blocks each, prior to transitioning into the outer H-grid far field. This is evident directly in the `blocks` entry of `blockMeshDict`:

1. **Inner O-grid (prism layer):** four blocks connecting the cylinder wall at R to the prism boundary at R_{prism}, resolved with $N_{bl} = 10$ cells in the radial direction and $N_c = 60$ cells circumferentially.
2. **Outer O-grid (transition layer):** four blocks connecting R_{prism} to the O-grid outer square at $R_{sq} = 1.5D$, resolved with N_{trans} cells radially.
3. **Outer H-grid (far field):** eight blocks connecting the O-grid outer square to the rectangular far-field boundary.

Because each block in the two O-grid rings is a topologically regular quadrilateral rather than a wedge-shaped sector, cell faces in the near-wall region remain close to orthogonal to the wall normal across the full circumference, avoiding the corner singularity characteristic of a single-block polar discretisation. The outer ring functions specifically as a buffer between the aggressively graded inner prism layer and the comparatively coarse far-field H-grid: without it, a single block would be required to accommodate the full expansion from the near-wall cell size to the far-field cell size, producing poor orthogonality and high skewness at the O-grid/H-grid corners. By interposing a transition block with its own independently specified expansion rate (E_{far}; Section 3.4), the total cell-size growth is distributed across two grading stages, and cell aspect ratio remains bounded throughout the near-body region.

The cylindrical wall and the prism-ring boundary are both defined as true circular arcs (OpenFOAM `arc` edge type with a specified mid-arc point), rather than being approximated by straight chords, so that the discretised wall geometry converges to the exact circular boundary as N_c is increased.

3.4 Mesh Grading Strategy

All blocks employ `simpleGrading`, which prescribes a constant cell expansion ratio along each local block direction. Two independent expansion rates are specified by the analyst—the near-wall growth rate E and the target far-field expansion ratio E_{far}—from which four derived grading scalars are computed and reused throughout `blockMeshDict` to ensure consistent transitions at every shared block interface. For a geometric sequence of N cells with constant ratio r between successive cell widths, OpenFOAM's `simpleGrading` parameter is the ratio of the final to the first cell width, r^{N-1}. The four derived gradings are summarised in Table 3.3.

Table 3.3: Mesh grading parameters, their governing expressions, and their evaluated values for the baseline configuration ($E = 1.5$, $N_{bl} = 10$, $E_{far} = 1.06$, $N_c = 60$).

Parameter	Expression	Value	Physical role
`BL_grading`	$E^{N_{bl}-1}$	38.44	Total expansion across the inner prism layer, directing cells to grow away from the wall.
`Trans_grading`	$E_{far}^{N_{trans}-1}$	3.82	Total expansion across the outer transition layer.
`G_xMax`, `G_yMax`	E_{far}^{N-1}	21.94, 10.29	Applied in far-field blocks whose local block coordinate increases in the direction of growth (downstream, outward-lateral).
`G_xMin`, `G_yMin`	$E_{far}^{-(N-1)}$	0.0770, 0.0972	Applied in far-field blocks whose local block coordinate increases *toward* the cylinder, inverting the grading so the physical cell size still decreases toward the O-grid interface.

3.5 Boundary Conditions

Five named boundary patches are defined in `blockMeshDict`; their geometric type is summarised in Table 3.4.

Table 3.4: Boundary patches and geometric types defined in `blockMeshDict`.

Patch	Mesh type	Notes
`cylinder`	`wall`	Composed of four faces (one per inner O-grid block) at each of Z_{min} and Z_{max}.
`inlet`	`patch`	Composed of three upstream H-grid faces spanning the full Y-extent at X_{min}; substituted with a case-specific velocity via `runCases.sh`.
`outlet`	`patch`	Composed of three downstream H-grid faces spanning the full Y-extent at X_{max}.
`topAndBottom`	`wall`	Composed of six faces (three upper, three lower) spanning the full X-extent at Y_{max} and Y_{min}.
`frontAndBack`	`wall`	Composed of all block faces at Z_{min} and Z_{max}; *not* assigned the `empty` type, confirming that the mesh is resolved as a genuine three-dimensional domain rather than a collapsed 2-D planar approximation.

The geometric types listed above are set exclusively within `blockMeshDict` and determine only the mesh connectivity and patch topology. The field-level boundary conditions actually applied during the flow solution (e.g. whether `topAndBottom` and `frontAndBack` are treated as no-slip walls, slip/symmetry surfaces, or otherwise; whether `outlet` uses a fixed-pressure or advective condition; the exact inlet turbulence specification) are defined in the case's `0/` field files, which were not among the files supplied for this chapter and are therefore not reported here. Only the substitution performed by `runCases.sh` is confirmed directly: the placeholder token `INLET_U` in `0/U` (and, where present, `0/include/turbulenceInit`) is replaced with the case-specific free-stream velocity U drawn from `cases.csv`.

3.6 Numerical Schemes and Solver Configuration

Rather than being fixed for all simulations, the temporal discretisation and solver settings are parameterised per case and injected by `runCases.sh` at case-generation time, drawing on dedicated columns of `cases.csv` (Table 3.5).

Table 3.5: Case-dependent solver and time-integration parameters, as substituted into `system/controlDict` and `system/fvSchemes` by `runCases.sh`.

Target file / entry	CSV column	Behaviour
`controlDict/application`	`solver`	Sets the OpenFOAM solver executable invoked for the case (e.g. a steady-state or transient incompressible solver, per case).
`controlDict/endTime`	`endTime`	Simulation end time or iteration count.
`controlDict/deltaT`	`dt`	Fixed time step (transient cases).
`controlDict/maxCo`	`maxCo`	Maximum permissible Courant number, used by adaptive time-stepping where enabled.
`fvSchemes/ddtSchemes`	`state`, `ddT`	If `state=steady`, the `ddtSchemes` block is overwritten with `steadyState`; otherwise it is overwritten with the transient scheme named in the `ddT` column.

Because the specific solver executable, transient scheme name, and associated convective/gradient/Laplacian discretisation and linear-solver settings are defined within the `baseCase` template's `system/fvSchemes` and `system/fvSolution` files—neither of which was supplied for this chapter—a scheme table of the form typically presented in this section (e.g. specifying the divergence, gradient, and Laplacian schemes, or the pressure-velocity coupling algorithm and under-relaxation factors) cannot be populated with verified values here. This should be completed directly from those files once available, following the same level of justification given for the mesh-generation parameters above.

3.7 Reference Quantities

The reference quantities used for non-dimensionalisation are defined in `geometryProperties` as

$$l_{\text{ref}} = D = 0.1\,\text{m}, \tag{3.2}$$

$$A_{\text{ref}} = 1.0 \cdot D \cdot D = 1.0 \times 10^{-2}\,\text{m}^2. \tag{3.3}$$

As with the solver configuration, the `forces` function-object dictionary that would integrate the pressure and viscous stress contributions over the `cylinder` patch to obtain C_D and C_L is defined in a `system/` file outside the set supplied for this chapter, and is therefore not reported here; only the two reference quantities feeding such a calculation are confirmed directly from `geometryProperties`.

3.8 Automated Parametric Campaign Execution

The full parameter set described above is exercised programmatically by `runCases.sh`, which reads `cases.csv` row by row and, for each entry, instantiates an independent case directory copied from a common `baseCase` template, applies the substitutions summarised in Table 3.5 together with the geometric-parameter substitutions (`D`, `N_bl`, `E`, `R_prism_mult`, `E_far`, `N_c`, `N_z`) via a pattern-anchored `sed` operation on `system/geometryProperties`, and then executes `blockMesh`, domain decomposition (`decomposePar`), the parallel solver (`mpirun`), and solution reconstruction (`reconstructPar`) under a common logging and error-handling framework, skipping any case whose output directory already exists. Because every derived mesh quantity discussed in Sections 3.3–3.4 is recomputed automatically by `blockMesh`'s evaluation of the `#calc` expressions whenever these seven parameters are substituted, the campaign automation guarantees topological and grading consistency across the full parametric sweep without case-specific manual intervention in the mesh definition.

4

Automated Simulation Campaign Framework

4.1 Motivation for Campaign-Level Automation

The parametric mesh-generation architecture described in Chapter 3 is designed so that every geometric and grading quantity in the mesh is a deterministic function of seven independent variables (D, N_c, N_z, N_{bl}, E, R_{prism_mult}, E_{far}). This design is only exploited fully, however, if the process of generating, meshing, and solving a large number of parameter combinations is itself automated and auditable; manually editing each case's dictionaries would reintroduce exactly the risk of inconsistency that the analytic parameterisation was designed to eliminate, and would not scale to a Reynolds-number sweep of the kind required for the validation and results chapters of this work. A dedicated driver script, `runCases.sh`, is therefore used to instantiate, configure, mesh, and execute every case in a campaign from a single tabulated specification, `cases.csv`, applying an identical, version-controlled procedure to every row and leaving a complete log trail for post-hoc verification.

4.2 Repository and Directory Architecture

The project root, `ROOT\_DIR`, is organised into a fixed set of top-level directories, each with a single, non-overlapping responsibility. This structure is established relative to the location of `runCases.sh` itself (via `SCRIPT\_DIR`), so that the campaign can be executed from any checkout of the repository without hard-coded absolute paths:

```
ROOT_DIR/
+-- baseCase/                    # Template case (never modified in place)
|   +-- 0/                       # Initial / boundary field values
|   |   +-- U
|   |   +-- epsilon
|   |   +-- k
|   |   +-- nut
|   |   +-- nuTilda
|   |   +-- omega
|   |   +-- p
```

```
|   |   +-- include/
|   |       +-- initialConditions
|   |       +-- turbulenceInit
|   +-- constant/
|   |   +-- transportProperties
|   |   +-- turbulenceProperties
|   +-- system/
|   |   +-- blockMeshDict
|   |   +-- geometryProperties
|   |   +-- controlDict
|   |   +-- fvSchemes
|   |   +-- fvSolution
|   |   +-- decomposeParDict
|   +-- Allclean.sh
|   +-- monitor_convergence.py
|   +-- yPlus.py
|   +-- foam.foam
+-- cases.csv                    # Campaign specification (one row per case)
+-- cases/                       # Instantiated, run cases (generated)
|   +-- case_Re100000/
|   +-- case_Re500000/
+-- logs/                        # Per-case execution logs (generated)
    +-- case_Re100000.log
    +-- case_Re500000.log
```

The `baseCase/` directory functions as a template: it is never edited in place during a campaign, and is instead copied wholesale into a fresh directory under `cases/` for every row of `cases.csv` (Section **??**), via `cp-r"\$BASEDIR""\$CASEDIR"`. This template-copy pattern, rather than an in-place-edit-and-restore pattern, guarantees that a failed or partially completed case can never corrupt the reference configuration used by subsequent cases, and that every instantiated case directory under `cases/` is a complete, independently runnable, and independently inspectable OpenFOAM case.

The presence of five turbulence-related field files in `0/` (`epsilon`, `k`, `nut`, `nuTilda`, `omega`), alongside the mean-flow fields `U` and `p`, indicates that the template case is provisioned for more than one turbulence-closure family (`k`, `epsilon`, and `nut` correspond to a k–ε-type or wall-function-based RANS closure; `omega` to a k–ω-type closure; `nuTilda` to the Spalart–Allmaras one-equation model), selected via `constant/turbulenceProperties`. `constant/transportProperties` correspondingly defines the working fluid's kinematic viscosity ν; the inlet velocities and Reynolds numbers appearing in `cases.csv` (Section 4.3) are mutually consistent with $\nu = 1.5 \times 10^{-5}\,\mathrm{m}^2/\mathrm{s}$, characteristic of air at standard conditions, for both sample cases inspected in this chapter. The two top-level utility scripts, `monitor_convergence.py` and `yPlus.py`, together with `Allclean.sh`, are copied into every case directory and are presumed to provide, respectively, residual/force convergence monitoring, post-run near-wall y^+ verification, and case cleanup; their internal logic was not among the files supplied for this thesis and is therefore not characterised further here. The exact contents of `fvSchemes`, `fvSolution`, `decomposeParDict`, `turbulenceProperties`, and `transportProperties` are likewise not verified in this chapter, for the same reason noted in Section 3.6 of

Chapter 3; only the entries that `runCases.sh` is confirmed to overwrite programmatically are reported with certainty.

4.3 Case Specification Schema

Every case in the campaign is specified by a single row of `cases.csv`, whose sixteen columns jointly determine the physical condition, temporal discretisation, and mesh-generation parameters for that case (Table 4.1).

Table 4.1: Column schema of `cases.csv` and the file or entry configured by each column.

Column	Target	Role
`Re`	Documentation only	Reynolds number of the case; not written into any dictionary directly.
`state`	`fvSchemes`	`steady` or `transient`; selects the `ddtSchemes` block.
`solver`	`controlDict/application`	OpenFOAM solver executable.
`ddT`	`fvSchemes/ddtSchemes`	Transient time-derivative scheme, used only when `state = transient`.
`U`	`0/U`, `0/include/turbulenceInit`	Free-stream inlet velocity; replaces `INLET_U`.
`endTime`	`controlDict/endTime`	Simulation end time.
`dt`	`controlDict/deltaT`	Fixed time step.
`maxCo`	`controlDict/maxCo`	Maximum Courant number.
`D`	`geometryProperties`	Cylinder diameter.
`N_bl`	`geometryProperties`	Inner O-grid (prism) layer count.
`E`	`geometryProperties`	Near-wall growth rate.
`R_prism_mult`	`geometryProperties`	Prism-block width multiplier.
`E_far`	`geometryProperties`	Target far-field expansion ratio.
`N_c`	`geometryProperties`	Circumferential resolution.
`N_z`	`geometryProperties`	Spanwise cell count.
`label`	`cases/<label>`, `logs/<label>.log`	Case directory and corresponding log-file name.

Two representative rows from the supplied campaign are reproduced in Table 4.2.

Table 4.2: Sample campaign rows from `cases.csv`.

Parameter	`case_Re100000`	`case_Re500000`
Re	100,000	500,000
State	transient	transient
Solver	`pimpleFoam`	`pimpleFoam`
ddt scheme	`backward`	`backward`
U (m s^{-1})	15	75
End time (s)	5	5
Δt (s)	1.25×10^{-3}	1.00×10^{-4}
$Co_{\max}$	10	10
D (m)	0.1	0.1
N_{bl}	5	5
E	1.5	1.5
$R_{\text{prism,mult}}$	0.6	0.6
E_{far}	1.07	1.07
N_c	40	40
N_z	10	10

Both rows imply $\nu = UD/Re = 1.5 \times 10^{-5}\,\text{m}^2/\text{s}$, so the two cases represent the same fluid and geometry at two different free-stream velocities, consistent with a Reynolds-number sweep at fixed D. Both specify the implicit second-order `backward` differencing scheme for the time derivative and a comparatively large maximum Courant number ($Co_{max} = 10$), consistent with the unconditionally-stable outer-corrector treatment of the PIMPLE algorithm implemented by `pimpleFoam`, which tolerates $Co > 1$ without a loss of stability (though possibly at some cost to temporal accuracy, which should be checked against a lower-Co case as part of the verification study). The correspondingly short fixed time step for the higher-Re case ($\Delta t = 1.00 \times 10^{-4}\,\text{s}$ versus $1.25 \times 10^{-3}\,\text{s}$) reflects the higher convective velocity at fixed cell size.

Finally, the CSV file supplied contains a wholly blank trailing row (all fields empty). This is handled explicitly and safely by `runCases.sh`, which skips any row for which either the `label` or `solver` field is empty, rather than allowing it to propagate into a malformed case directory.

4.4 Execution Pipeline and Fault Tolerance

Once a case directory has been fully configured, `runCases.sh` executes a fixed four-stage pipeline within that directory, each stage gated on the success of the previous one:

1. `blockMesh` — generates the block-structured mesh described in Chapter 3 from the now-populated `geometryProperties` and `blockMeshDict`.
2. `decomposePar -force` — decomposes the domain into `$PARALLEL_JOBS` sub-domains (eight, by default; configurable via the `--parallel` command-line flag) according to `system/decomposeParDict`.

3. `mpirun -np \$PARALLEL_JOBS <solver> -parallel` — executes the case-specified solver (`pimpleFoam` in both sample cases) in parallel.

4. `reconstructPar` — reassembles the per-processor solution fields into a single serial time-directory structure, after which the intermediate `processor*` directories are deleted to limit disk usage.

4.5 Logging and Campaign Traceability

Every case produces an independent, append-only log file at `logs/<label>.log`, into which every substitution step and every pipeline stage (Section 4.4) writes a human-readable status line via `tee -a`, alongside the full stdout/stderr of `blockMesh`, `decomposePar`, the solver, and `reconstructPar`. This provides a complete, per-case audit trail that can be inspected independently of the case directory itself, which is useful both for debugging a failed case and for archiving campaign provenance alongside the results reported in later chapters.

At the conclusion of the campaign, the script aggregates the outcome of every attempted case into two arrays, `COMPLETED_CASES` and `FAILED_CASES`, and prints a consolidated summary reporting the total number of cases completed and failed, together with the specific failure stage for each unsuccessful case. This summary, combined with the per-case logs, allows the full campaign — comprising an arbitrary number of rows in `cases.csv` — to be re-run unchanged: because already-existing case directories are skipped, re-invoking `runCases.sh` after addressing a failure cause will re-attempt only the cases that previously failed or were never started, without disturbing successfully completed results.

5

Results and Numerical Assessment

This chapter summarizes the numerical results obtained from the cylinder-flow simulations and discusses the main observations related to the numerical schemes, solver selection, and simulation parameters. The study was initially performed at low Reynolds numbers to establish a validated baseline, followed by investigations of Reynolds-number effects, temporal discretization, and solver selection.

5.1 Low-Reynolds-Number Validation

The initial simulations were performed at a low Reynolds number to verify the cylinder geometry, boundary conditions, fluid properties, and calculation of the drag coefficient.

For the first validation case, the cylinder diameter was set to $D = 0.01\,\mathrm{m}$ and the inlet velocity to $U = 0.01\,\mathrm{m/s}$. The fluid properties were $\rho = 998\,\mathrm{kg/m^3}$, $\mu = 0.001002\,\mathrm{Pa\,s}$, and $\nu = 1.004 \times 10^{-6}\,\mathrm{m^2/s}$. The resulting Reynolds number was

$$Re = \frac{UD}{\nu} \approx 99.6. \tag{5.1}$$

For very low Reynolds numbers, the flow around a cylinder is characterized by creeping flow. At higher Reynolds numbers within the laminar regime, analytical and empirical correlations can be used as reference values for the drag coefficient.

For the present case, the reference drag coefficient was $C_d = 1.0938$. The corresponding CFD result was $C_d = 1.1564 \pm 0.0025$, giving an absolute error of 0.0626 and a relative error of 5.73%.

The simulated drag coefficient is approximately 5.7% higher than the reference value. The small variation of the calculated drag coefficient indicates a stable result for this case. However, the remaining difference demonstrates that agreement with a reference value depends on the mesh, numerical schemes, boundary conditions, and other numerical settings.

Learning: A low-Reynolds-number validation case provides a useful first step for checking the numerical setup and force-coefficient calculation before proceeding to more demanding simulations.

Table 5.1: Validation of the low-Reynolds-number cylinder case at $Re \approx 99.6$.

Quantity	Value
Cylinder diameter, D	$0.01\,\mathrm{m}$
Inlet velocity, U	$0.01\,\mathrm{m/s}$
Density, ρ	$998\,\mathrm{kg/m^3}$
Dynamic viscosity, μ	$1.002 \times 10^{-3}\,\mathrm{Pa\,s}$
Kinematic viscosity, ν	$1.004 \times 10^{-6}\,\mathrm{m^2/s}$
Reynolds number, Re	99.6
Reference C_d	1.0938
CFD C_d	1.1564 ± 0.0025
Absolute error	0.0626
Relative error	5.73%

5.2 Reynolds-Number and Numerical-Scheme Study

A systematic study was subsequently performed at Reynolds numbers of $Re = 20$, 40, and 100. Steady simulations were performed using `simpleFoam`, while transient simulations were performed using `pisoFoam`. Different temporal discretization schemes were investigated for the $Re = 100$ cases.

Table 5.2: Comparison of drag coefficients for different Reynolds numbers and numerical schemes.

Case	Re	C_d	Reference	Error (%)
Re20_steady	20	2.0520	2.2200	7.6
Re40_steady	40	1.5037	1.5000	0.2
Re100_Euler	100	1.2968	1.3500	3.9
Re100_backward	100	1.2796	1.3500	5.2
Re100_CN_0.9	100	1.2576	1.3500	6.8

The results show that the predicted drag coefficient depends on both Reynolds number and numerical formulation. The $Re = 40$ steady case shows the smallest error in this group, with a difference of only 0.2% from the selected reference value.

For the $Re = 100$ cases, the Euler scheme gives the smallest error among the three initially tested temporal schemes. The predicted drag coefficient decreases from 1.2968 using Euler discretization to 1.2576 using Crank–Nicolson with a coefficient of 0.9.

This indicates that the temporal discretization can have a noticeable influence on the predicted drag coefficient. Therefore, the numerical scheme should be considered as part of the verification process rather than treated only as a numerical implementation detail.

Learning: Numerical schemes should be kept consistent when comparing physical parameters. If the objective is to compare different numerical schemes, only the intended numerical parameter should be changed while the remaining simulation setup is kept unchanged.

5.3 Crank–Nicolson Sensitivity at $Re = 100$

A further study was performed at $Re = 100$ using the Crank–Nicolson scheme with different blending coefficients. The objective was to investigate the sensitivity of the predicted drag coefficient to this numerical parameter.

Table 5.3: Sensitivity of the drag coefficient to the Crank–Nicolson coefficient at $Re = 100$.

Case	CN coefficient	C_d	Error (%)
Re100_CN_0.2	0.2	1.2454	7.7
Re100_CN_0.4	0.4	1.2304	8.9
Re100_CN_0.5	0.5	1.2230	9.4
Re100_CN_0.6	0.6	1.2161	9.9
Re100_CN_0.8	0.8	1.2034	10.9

A systematic decrease in the predicted drag coefficient is observed as the Crank–Nicolson coefficient increases. Consequently, the relative error increases from 7.7% at a coefficient of 0.2 to 10.9% at a coefficient of 0.8.

The results demonstrate that the Crank–Nicolson coefficient has a measurable influence on the calculated drag coefficient for the present setup. The coefficient should therefore be documented and considered during numerical sensitivity studies.

Learning: The Crank–Nicolson coefficient should not be changed without assessing its effect on the quantities of interest. For reproducible simulations, the selected value should always be recorded together with the other numerical settings.

5.4 PISO and PIMPLE Comparison

The next stage of the study investigated the use of `pimpleFoam` and compared its results with the previously obtained transient simulations. The comparison was initially performed at $Re = 100$, followed by an additional simulation at $Re = 200$.

Table 5.4: Results obtained using `pimpleFoam` at $Re = 100$ and $Re = 200$.

Case	Re	C_d	Reference	Error (%)
Re100_PIMPLE_Euler	100	1.3002	1.3500	3.7
Re100_PIMPLE_backward	100	1.3850	1.3500	2.6
Re100_PIMPLE_CN_0.9	100	1.3782	1.3500	2.1
Re200_PIMPLE_backward	200	1.4365	1.3300	8.0

For the $Re = 100$ PIMPLE cases, all three temporal discretization schemes produced results closer to the reference value than the corresponding cases from the preceding PISO study.

The Crank–Nicolson case with a coefficient of 0.9 produced the smallest error among the tested PIMPLE cases, with an error of 2.1%. The backward scheme produced an error of 2.6%, while the Euler scheme produced an error of 3.7%.

The $Re = 200$ PIMPLE case produced $C_d = 1.4365$, compared with the reference value of 1.3300, corresponding to an error of 8.0%. The increased deviation indicates that additional verification is required when extending the numerical setup to higher Reynolds numbers.

5.4.1 Interpretation of PISO and PIMPLE Results

For the cases investigated in this study, the PIMPLE simulations at $Re = 100$ produced drag coefficients that were closer to the selected reference value than the corresponding PISO simulations. However, this result alone is not sufficient to conclude that PIMPLE is generally more accurate than PISO.

The two solvers use different pressure–velocity coupling strategies and may respond differently to the selected time step and numerical settings. A complete comparison should also consider residual convergence, Courant-number behaviour, computational cost, time-step sensitivity, force-coefficient stability, and mass conservation.

Based on the simulations performed so far, `pimpleFoam` provides a suitable framework for the transient cases investigated here. Further testing is required before making a general solver-selection recommendation.

5.5 Grid, Time-Step, and Parallelization Sensitivity

After the initial solver and numerical-scheme investigations, a systematic grid and time-step sensitivity study was performed using `pimpleFoam` at $Re = 200$. The objective was to determine whether the calculated aerodynamic quantities were sufficiently independent of the spatial and temporal resolution.

The simulations were performed using a parameterized mesh, with the grid resolution represented by the number of cells in the streamwise and transverse directions. The cases were labelled G1, G2, G3, and G4 for increasing spatial resolution. Additional simulations with different time steps were labelled T1 to T4.

5.5.1 Grid Sensitivity

The grid sensitivity study was performed using a constant time step of $\Delta t = 0.001\,\mathrm{s}$. The grid resolution was increased from 20×20 cells in the main structured region for G1 to 100×100 for G4.

Table 5.5: Grid sensitivity study at $Re = 200$ using $\Delta t = 0.001\,\mathrm{s}$.

Case	N_x	N_y	**Cells**	C_d	$C_{l,\mathrm{rms}}$	**Exec. time (s)**
G1	20	20	6,000	1.3167	0.4070	195.79
G2	40	40	24,000	1.3834	0.4992	448.77
G3	80	80	96,000	1.3771	0.5153	1571.06
G4	100	100	150,000	1.3727	0.5231	3370.34

The results show a clear change between the coarse G1 mesh and the finer meshes. The mean drag coefficient increases from $C_d = 1.3167$ for G1 to 1.3834 for G2. Further

refinement to G3 changes the drag coefficient to 1.3771, corresponding to a difference of approximately 0.46% relative to G2.

A further refinement from G3 to G4 changes the drag coefficient from 1.3771 to 1.3727, corresponding to a difference of approximately 0.32%. The relatively small change between G3 and G4 indicates that the drag coefficient is becoming insensitive to further spatial refinement.

The same trend can be observed for the lift fluctuations. The value of $C_{l,\mathrm{rms}}$ increases from 0.4070 for G1 to approximately 0.52 for the finer grids. The change becomes considerably smaller between G3 and G4, where the values are 0.5153 and 0.5231, respectively.

The results therefore indicate that G1 is too coarse to provide a sufficiently grid-independent solution, while G3 already provides a relatively stable prediction of the main force coefficients. G4 provides only a small additional change in C_d at a substantially increased computational cost.

The computational cost also increases strongly with grid resolution. Increasing the number of cells from approximately $6,000$ in G1 to $150,000$ in G4 increases the execution time from approximately 196 s to 3370 s. This demonstrates that further mesh refinement should be justified by a corresponding improvement in the quantities of interest.

Learning: The grid-sensitivity results indicate that the G3 mesh provides a suitable balance between spatial resolution and computational cost for the present $Re = 200$ study. The change from G3 to G4 is small for the mean drag coefficient, suggesting that further refinement provides limited benefit for this quantity.

5.5.2 Parallelization Study

The effect of parallel execution was also examined using the G3 mesh. The same case was executed using four and six processor cores.

Table 5.6: Parallel execution comparison for the G3 mesh at $Re = 200$.

Case	Cores	Cells	Clock time (s)	Execution time (s)
G3_4Core	4	96,000	4893	2782.79
G3_6Core	6	96,000	3532	1571.06

Increasing the number of processor cores from four to six reduced the measured execution time from approximately 2783 s to 1571 s. This corresponds to a reduction of approximately 44% in execution time.

The scaling is not perfectly proportional to the increase in processor count, which is expected because of parallel communication, decomposition, reconstruction, and other computational overheads. Nevertheless, the results demonstrate a clear benefit from parallel execution for the present mesh.

The G3 case therefore provides not only a suitable spatial resolution but also a practical computational size for parallel simulations.

Learning: Parallelization can significantly reduce computational time, but the optimum processor count should be determined based on the mesh size and scaling behaviour rather than simply increasing the number of cores.

5.5.3 Time-Step Sensitivity

A time-step sensitivity study was subsequently performed using the G3 mesh with $N_x = N_y = 80$ and 96,000 cells. The physical simulation time was kept consistent while the time step was progressively reduced from 0.002 s to 0.00025 s.

Table 5.7: Time-step sensitivity study for the G3 mesh at $Re = 200$.

Case	Δt (s)	Timesteps	C_d	$C_{d,\mathrm{std}}$	$C_{l,\mathrm{rms}}$
T1	0.00200	500	1.4024	0.0500	0.5698
T2	0.00100	1000	1.3756	0.0708	0.5145
T3	0.00050	2000	1.3630	0.0806	0.4914
T4	0.00025	4000	1.3517	0.0964	0.4801

A systematic reduction in the predicted mean drag coefficient is observed as the time step is reduced. The coarsest time step, T1, gives $C_d = 1.4024$, while the finest time step, T4, gives $C_d = 1.3517$.

The change between successive time-step levels decreases as the time step becomes smaller. The change in C_d from T1 to T2 is approximately 1.9%, while the change from T2 to T3 is approximately 0.9%. The change from T3 to T4 is approximately 0.8%.

This indicates that the solution is approaching temporal convergence as the time step is reduced. The difference between T3 and T4 is relatively small compared with the difference between the coarser time steps. Therefore, T3 provides a reasonable compromise between temporal accuracy and computational cost, while T4 can be used when a more conservative temporal-resolution check is required.

The effect on the lift fluctuations is also visible. The calculated $C_{l,\mathrm{rms}}$ decreases from 0.5698 for T1 to 0.4801 for T4. This indicates that the time step influences not only the mean drag but also the predicted unsteady force behaviour.

The computational cost increases substantially with temporal refinement. Reducing the time step from 0.001 s to 0.00025 s increases the number of time steps from 1000 to 4000 and increases the execution time from approximately 1979 s to 6047 s.

Learning: The time step should be selected based on temporal sensitivity rather than computational convenience alone. For the present study, the results from T3 and T4 are relatively close, indicating that $\Delta t = 0.0005$ s provides a reasonable starting point for further simulations, while $\Delta t = 0.00025$ s provides a more conservative temporal-resolution check.

5.5.4 Combined Grid and Time-Step Assessment

The grid and time-step studies demonstrate that both spatial and temporal resolution influence the calculated force coefficients. The grid study showed that refinement beyond G3 produces only a small change in the mean drag coefficient, whereas the time-step study showed a systematic change in C_d as Δt was reduced.

For the present $Re = 200$ configuration, the combination of the G3 mesh and $\Delta t = 0.0005$ s provides a practical compromise between accuracy and computational cost. The finest cases can be retained as verification cases to confirm that the selected production setup remains sufficiently resolved.

It is important to note that grid and time-step independence should be assessed together with the relevant physical quantities. A simulation should not be considered fully verified solely because the change in mean C_d is small. The behaviour of C_l, vortex shedding, residuals, mass conservation, and other quantities of interest should also be considered.

Learning: The sensitivity studies provide a basis for selecting simulation parameters systematically rather than relying on arbitrary mesh density or time-step values. The selected production configuration should therefore be justified using both accuracy and computational cost.

5.6 Turbulence Modelling

The previous simulations were performed using a laminar flow assumption. The next stage of the study introduced the standard k-epsilon turbulence model. The objective was to establish consistent inlet turbulence conditions and to investigate the resulting aerodynamic coefficients and wall resolution.

5.6.1 Inlet Turbulence Specification

A turbulence intensity of 1

$$L = 0.07D \tag{5.2}$$

The turbulent kinetic energy was calculated from the inlet velocity using

$$k_0 = 1.5I^2U_{\text{inlet}}^2 \tag{5.3}$$

where I is the turbulence intensity and U_{inlet} is the inlet velocity.

The turbulent dissipation rate was calculated using

$$\epsilon_0 = \frac{C_\mu^{3/4}k_0^{3/2}}{L} \tag{5.4}$$

where L is the turbulence length scale and $C_\mu = 0.09$ is the standard model constant.

The inlet turbulence quantities were therefore calculated automatically from the inlet velocity, providing consistent turbulence conditions when changing the Reynolds number.

5.6.2 Simulation Configuration

Three Reynolds numbers were investigated using the transient solver `pimpleFoam`. The standard k-epsilon turbulence model was used for all cases. All simulations used an 80 by 80 mesh and the backward time discretisation scheme.

The mean drag coefficient, root-mean-square lift coefficient, maximum wall y-plus, and mean wall y-plus were extracted from each simulation.

5.6.3 Results

The results obtained using the standard k-epsilon model are given in Table 5.9.

Table 5.8: Simulation parameters for the turbulence model study.

Case	Re	U	End	dt	Nx	Ny
Re200_kEpsilon	200	0.30	10	0.0010	80	80
Re500_kEpsilon	500	0.75	10	0.0005	80	80
Re1000_kEpsilon	1000	1.50	10	0.0001	80	80

Table 5.9: Results from the standard k-epsilon simulations.

Case	Re	Cd	Cl rms	Max y+	Mean y+
Re200_kEpsilon	200	1.2914	0.6457	1.5104	0.2402
Re500_kEpsilon	500	1.2613	0.6308	2.9413	0.4794
Re1000_kEpsilon	1000	1.1421	0.5711	4.5698	0.7542

The mean drag coefficient decreases with increasing Reynolds number. The predicted value decreases from 1.2914 at Reynolds number 200 to 1.1421 at Reynolds number 1000.

The lift fluctuations also decrease with increasing Reynolds number. The root-mean-square lift coefficient decreases from 0.6457 at Reynolds number 200 to 0.5711 at Reynolds number 1000.

In contrast, the wall y-plus values increase with Reynolds number. The maximum value increases from 1.5104 at Reynolds number 200 to 4.5698 at Reynolds number 1000. The corresponding mean value increases from 0.2402 to 0.7542.

5.6.4 Initial Assessment

The simulations show that the wall y-plus values remain relatively low for all three Reynolds numbers considered. The maximum value remains below 5, although a systematic increase is observed with increasing Reynolds number.

The increase in wall y-plus indicates that wall resolution should continue to be monitored when extending the simulations to higher Reynolds numbers. The suitability of the near-wall treatment should also be considered together with the selected turbulence model and mesh resolution.

These simulations represent an initial investigation of the standard k-epsilon model rather than a complete turbulence-model validation. A complete assessment would require comparison with appropriate reference data together with residual convergence, force-coefficient convergence, mesh sensitivity, and time-step sensitivity.

A subsequent comparison with the k-omega SST model can be used to investigate the sensitivity of the predicted flow quantities to the turbulence-model formulation.

5.7 Turbulence Modelling: k-omega SST Model

Following the initial investigation using the standard k-epsilon model, the k-omega SST turbulence model was investigated at higher Reynolds numbers. The objective was to examine the behaviour of the flow predictions as the Reynolds number was increased and to compare the results with those obtained using the standard k-epsilon model.

5.7.1 Simulation Configuration

The k-omega SST simulations were performed using the transient solver `pimpleFoam`. An 80 by 80 mesh was used for all cases. The end time was set to 1.5 s. A time step of 0.0001 s was used up to Reynolds number 3900, while a smaller time step of 0.00005 s was used for the Reynolds number 4500 case.

Table 5.10: Simulation parameters for the k-omega SST study.

Re	U	End	dt	Nx	Ny
1000	1.50	1.5	0.00010	80	80
1500	2.25	1.5	0.00010	80	80
2000	3.00	1.5	0.00010	80	80
2500	3.75	1.5	0.00010	80	80
3000	4.50	1.5	0.00010	80	80
3500	5.25	1.5	0.00010	80	80
3900	5.85	1.5	0.00010	80	80
4500	6.75	1.5	0.00005	80	80

5.7.2 Results and Comparison

The k-omega SST simulations produced drag coefficients in the approximate range of 1.51 to 1.71 over the Reynolds number range investigated. These values were higher than those obtained using the standard k-epsilon model, for which the corresponding drag coefficients were approximately 1.13 to 1.23.

The SST results also showed an increase in the predicted drag coefficient with increasing Reynolds number. This trend is different from the expected behaviour of the experimental data considered in this study, which show an approximately constant or slightly decreasing drag coefficient over the corresponding Reynolds number range.

The predicted Strouhal numbers were approximately 0.44 to 0.49. These values are substantially higher than the expected experimental range of approximately 0.20 to 0.25. The simulations therefore predicted a vortex shedding frequency approximately twice as high as the experimental value.

5.7.3 Flow Behaviour

Inspection of the flow field indicated that the simulated wake remained dominated by coherent two-dimensional vortical structures. No significant spanwise breakup of the vortices was observed in the present simulations. This may contribute to an artificially coherent wake and consequently higher predicted aerodynamic forces.

The k-omega SST model also provides stronger near-wall treatment than the standard k-epsilon model. In the present simulations, this was associated with higher predicted wall shear and consequently higher drag coefficients. However, the difference between the two turbulence models should not be attributed solely to the turbulence-model formulation. The two-dimensional mesh, absence of spanwise vortex breakup, wall resolution, and temporal resolution can all influence the predicted forces.

5.7.4 Wall Resolution

The maximum wall y-plus increased with Reynolds number, from approximately 4.6 at the lower end of the investigated range to approximately 10 at the highest Reynolds number.

The increasing y-plus indicates that the near-wall resolution becomes progressively coarser in wall units as the Reynolds number increases. Although the values remain relatively low compared with many wall-function applications, the near-wall resolution should be monitored when interpreting the SST results because the model is sensitive to the treatment of the boundary layer.

5.7.5 Assessment of the SST Results

The k-omega SST simulations provided a useful comparison with the standard k-epsilon calculations and demonstrated a strong sensitivity of the predicted drag and vortex shedding behaviour to the turbulence modelling approach.

However, the increasing drag coefficient with Reynolds number and the predicted Strouhal numbers of approximately 0.44 to 0.49 indicate that the present setup does not reproduce the expected experimental behaviour satisfactorily over the investigated range.

A major limitation is that the simulations are effectively two-dimensional. At higher Reynolds numbers, three-dimensional wake structures and spanwise vortex breakup can become important. Their absence can maintain artificially coherent vortices and influence both the mean drag and shedding frequency.

Therefore, the SST results should be considered as a numerical sensitivity study rather than a validated prediction of the high-Reynolds-number flow. Further improvement would require investigation of three-dimensional modelling, mesh refinement, time-step sensitivity, near-wall resolution, and comparison with experimental force and shedding-frequency data.

5.8 Extension to Higher Reynolds Numbers and Three-Dimensional Modelling

The preceding simulations demonstrated that the present two-dimensional RANS approach can reproduce the general behaviour of the cylinder flow at lower Reynolds numbers, but increasing discrepancies become apparent as the Reynolds number is increased. In particular, the predicted drag coefficient and Strouhal number show a non-linear variation with Reynolds number in the higher-Reynolds-number simulations. This behaviour indicates that the present modelling approach does not capture all of the relevant physical flow structures at higher Reynolds numbers.

The limitation becomes particularly important when extending the study towards Reynolds numbers in the range of approximately 10^5 to 10^7. In this regime, the flow contains increasingly complex turbulent structures, strong three-dimensional effects, and changes in the boundary-layer and wake behaviour. A two-dimensional model cannot represent spanwise vortex deformation, vortex dislocation, and three-dimensional wake instabilities. Consequently, maintaining a two-dimensional coherent vortex struc-

ture can lead to unrealistic predictions of the mean drag and vortex shedding frequency.

The comparison between the standard k-epsilon and k-omega SST simulations also demonstrated that changing the RANS turbulence model alone does not remove these discrepancies. Although the two models produce different drag and lift predictions, both exhibit limitations in reproducing the expected Reynolds-number dependence of the aerodynamic coefficients. This suggests that the modelling limitations are not solely associated with the choice of RANS turbulence model.

5.8.1 Recommended Three-Dimensional Extension

The next stage of the study should therefore extend the computational domain to three dimensions. A three-dimensional cylinder geometry would allow the development of spanwise flow structures and three-dimensional wake instabilities that are absent from the current simulations.

The first step should be a three-dimensional RANS calculation using a suitably refined mesh. This would provide a direct comparison with the existing two-dimensional results while allowing the influence of the spanwise direction to be isolated.

Particular attention should be given to the spanwise domain length, mesh resolution in the wake, near-wall resolution, and time-step size. The resulting drag coefficient, lift fluctuations, Strouhal number, residuals, mass balance, and wall y-plus should be monitored consistently with the previous simulations.

5.8.2 Potential Application of LES

Large Eddy Simulation (LES) represents a possible next step for resolving the unsteady turbulent structures that become increasingly important at high Reynolds numbers. Unlike the present RANS simulations, LES resolves the larger turbulent structures directly while modelling the smaller scales.

LES could therefore provide a more detailed representation of the three-dimensional wake and vortex-shedding process. It would also provide a useful reference for assessing whether the non-linear behaviour observed in the present RANS calculations originates from the turbulence model, the two-dimensional assumption, or other numerical effects.

However, LES at Reynolds numbers between 10^5 and 10^7 can be computationally expensive. A sufficiently resolved near-wall LES would require a substantially finer mesh and smaller time step than the present simulations. Therefore, a staged approach is recommended rather than immediately applying LES to the highest Reynolds numbers.

A practical progression would be:

1. Extend the cylinder model from two dimensions to three dimensions.
2. Perform a three-dimensional RANS calculation at a moderate Reynolds number.
3. Verify mesh, time-step, wall resolution, and statistical convergence.
4. Compare the three-dimensional RANS results with the existing two-dimensional results.

5. Perform a three-dimensional LES at a selected Reynolds number where reliable reference data are available.

6. Validate the predicted mean drag coefficient and Strouhal number against experimental data.

7. Only after validation, extend the methodology towards the higher Reynolds-number range of 10^5 to 10^7.

The primary objective of this extension should not be to obtain a single drag coefficient, but to establish a numerical methodology that reproduces the correct physical trends with Reynolds number. In particular, the dependence of C_d, the Strouhal number, wake structure, and force fluctuations should be assessed together.

5.8.3 Recommended Validation Strategy

For the continuation of the project, the Reynolds-number study should be treated as a validation problem rather than only a parameter study. Experimental or established benchmark data should be used at selected Reynolds numbers covering the transition from the current RANS regime towards the intended high-Reynolds-number range.

The comparison should include the mean drag coefficient, root-mean-square lift coefficient, Strouhal number, and, where available, the qualitative wake structure. Mesh and time-step sensitivity should be demonstrated for the selected validation cases.

The present results therefore provide a useful baseline for the next stage of the project. The existing automated workflow, case-generation scripts, convergence monitoring, post-processing tools, and sensitivity studies can be retained and extended to three-dimensional RANS and subsequently LES. This provides a reproducible starting point for investigating high-Reynolds-number cylinder flow while avoiding the assumption that a change of turbulence model alone is sufficient to recover the missing physical behaviour.

6

Conclusions and Future Work

6.1 Summary of the Work

This project developed and investigated a fully parameterised, block-structured meshing and case-automation framework for the computational study of incompressible flow past a circular cylinder in OpenFOAM. The framework was designed to provide a reproducible basis for systematic numerical studies in which mesh resolution, time step, Reynolds number, solver settings, and turbulence modelling can be varied without manually rebuilding individual cases.

The meshing framework uses a nested double O-grid topology. An inner prism ring provides the near-wall resolution, while an outer transition ring connects the near-cylinder region to the far-field H-grid. The geometric and grading quantities are generated from a set of independent parameters, including the cylinder diameter, circumferential and radial cell counts, boundary-layer cell count, expansion ratios, and prism-layer extent. The parameterisation is implemented through OpenFOAM's `#calc` functionality and the `geometryProperties` dictionary.

The far-field cell counts and grading are derived from the matched interface cell width, providing continuity of cell size between the O-grid and H-grid regions. This avoids abrupt changes in cell size at internal block interfaces and allows mesh refinement to be performed systematically.

The campaign-automation layer uses `cases.csv` together with `runCases.sh` to generate, mesh, solve, and post-process multiple cases. The reference `baseCase` is retained as a template, while individual cases are generated from the parameter table. The execution pipeline consists of `blockMesh`, optional domain decomposition, parallel solver execution, and reconstruction. Per-case logs and automated post-processing provide traceability across the simulation campaign.

Following development of the framework, the methodology was applied to a series of cylinder-flow simulations. The numerical investigation progressed from low-Reynolds-number validation and solver comparisons to grid and time-step sensitivity studies, parallelisation tests, and turbulence-model investigations.

6.2 Numerical Results and Main Conclusions

The initial low-Reynolds-number calculations provided a baseline assessment of the cylinder-flow setup. The initial validation case produced a relative drag-coefficient er-

ror of approximately 5.73%, while the $Re = 40$ steady case showed an error of approximately 0.2% relative to the selected reference value. These cases demonstrated that the basic geometry, boundary conditions, force extraction, and numerical workflow were capable of producing physically reasonable results.

At $Re = 100$, the investigation showed that the numerical formulation has a measurable influence on the predicted drag coefficient. In particular, changes in the Crank–Nicolson coefficient resulted in a systematic change in the predicted C_d. The tested PIMPLE cases also produced smaller errors than the corresponding PISO cases considered in this study. This should not be interpreted as a general statement that PIMPLE is universally more accurate than PISO; rather, it demonstrates that solver and temporal formulation should be assessed together with the particular flow and numerical configuration.

The grid-sensitivity study at $Re = 200$ demonstrated a clear influence of spatial resolution on the predicted aerodynamic quantities. The drag coefficient increased from approximately 1.32 on the coarsest grid to approximately 1.38 on the finer grids. The difference between the G3 and G4 meshes was comparatively small, with the drag coefficient changing by less than one percent. The G3 mesh therefore provided a practical compromise between spatial resolution and computational cost for the investigated configuration, although a formal grid-convergence or GCI analysis was not performed.

The time-step study at $Re = 200$ also demonstrated temporal sensitivity. Reducing the time step from 0.002 s to 0.00025 s resulted in a systematic decrease in the predicted drag coefficient, from approximately 1.40 to 1.35. The changes became smaller as the time step was reduced, indicating an approach towards temporal convergence. The computational cost increased substantially with decreasing time step, demonstrating the need to balance temporal accuracy against computational expense.

The parallelisation study showed that increasing the number of processor cores from four to six reduced the execution time of the tested G3 case by approximately 44%. The corresponding speed-up was approximately 1.77. The scaling was therefore sublinear, as expected from communication, decomposition, and reconstruction overhead. The aerodynamic coefficients from the parallel runs remained close, indicating that the observed difference in execution time was not accompanied by a major change in the calculated mean flow quantities.

6.3 Turbulence Modelling Results

The standard k-epsilon model was subsequently introduced to investigate turbulent-flow behaviour. Simulations at $Re = 200$, 500, and 1000 showed a decrease in the predicted mean drag coefficient from approximately 1.29 to 1.14 as the Reynolds number increased. The root-mean-square lift coefficient also decreased over this range.

The wall-resolution assessment showed that the maximum y^+ increased from approximately 1.5 at $Re = 200$ to approximately 4.6 at $Re = 1000$. The values remained relatively low for the investigated cases, but the systematic increase demonstrates the importance of monitoring near-wall resolution as the Reynolds number is increased.

The k-omega SST model was subsequently applied to a higher Reynolds-number range from $Re = 1000$ to 4500. The SST simulations produced substantially higher drag coefficients than the standard k-epsilon simulations and predicted Strouhal numbers of approximately 0.44–0.49. The predicted Strouhal numbers were significantly higher

than the expected experimental range considered in this study.

The SST simulations also showed a non-linear variation of the aerodynamic quantities with Reynolds number. In particular, the predicted drag coefficient increased over the investigated range, whereas the available experimental behaviour indicates an approximately constant or slightly decreasing drag coefficient over the corresponding range.

These results demonstrate that changing the RANS turbulence model alone does not resolve all of the discrepancies observed at higher Reynolds numbers. The predicted behaviour is influenced not only by the turbulence closure, but also by the two-dimensional modelling assumption, wake structure, near-wall resolution, spatial resolution, and temporal resolution.

6.4 Limitations

Several limitations of the present study should be considered when interpreting the results.

The near-wall mesh spacing is currently controlled through geometric parameters rather than being generated automatically from a prescribed Reynolds-number-dependent y^+ target. Consequently, the resulting y^+ must be checked after each simulation.

The present study is primarily two-dimensional. This is a significant limitation for higher-Reynolds-number cylinder flow because three-dimensional wake structures and spanwise instabilities cannot develop in a two-dimensional formulation. The resulting wake can therefore remain more coherent than the physical flow.

The high-Reynolds-number simulations also demonstrate that the present RANS approach does not reproduce all experimentally observed trends in drag and vortex shedding. The turbulence-model comparison indicates that the discrepancy cannot be attributed solely to the choice between the k-epsilon and k-omega SST models.

The grid and time-step investigations provide evidence of sensitivity, but they do not constitute a formal uncertainty-quantification procedure. Likewise, the available validation cases cover only part of the intended Reynolds-number range.

The blockage ratio, domain extent, near-wall resolution, numerical schemes, and turbulence-model assumptions should therefore be considered when extending the work to substantially higher Reynolds numbers.

6.5 Future Work

The most immediate continuation of the study should be the extension of the present two-dimensional model to three dimensions. A three-dimensional cylinder model would allow spanwise flow structures, vortex deformation, and three-dimensional wake instabilities to develop.

The first stage should be a three-dimensional RANS calculation at a moderate Reynolds number for which reliable experimental reference data are available. The three-dimensional results should be compared directly with the existing two-dimensional calculations to quantify the influence of the additional spatial dimension.

The near-wall mesh-generation procedure should also be improved by introducing

an automated y^+-based first-cell-height calculation. The target y^+ should be selected according to the intended turbulence model and wall treatment, with the corresponding first-cell height determined from an appropriate estimate of the wall shear stress.

The far-field domain dimensions should be exposed as independent parameters in `geometryProperties` and `cases.csv`. This would allow domain-sensitivity studies to be performed with the same automated workflow as the existing mesh and time-step studies.

For higher Reynolds numbers, three-dimensional LES represents a potential next step. LES would allow the larger turbulent and unsteady wake structures to be resolved directly rather than represented entirely through a RANS closure. However, LES at Reynolds numbers approaching 10^5–10^7 can require substantially greater spatial and temporal resolution and computational resources.

A staged approach is therefore recommended. Three-dimensional RANS should first be used to establish the influence of the spanwise direction. LES can then be performed at selected Reynolds numbers for which reliable experimental data are available. The LES methodology should be validated against mean drag, lift fluctuations, Strouhal number, and, where possible, qualitative wake characteristics before being extended towards the highest Reynolds numbers.

The existing automation framework provides a suitable basis for this extension. The `cases.csv` approach can be expanded to include three-dimensional mesh parameters, spanwise domain size, turbulence-model selection, and LES-specific numerical parameters without fundamentally changing the campaign-management structure.

6.6 Recommended Best Practices

Based on the numerical investigations performed, the following practices are recommended for future simulations:

1. Begin with a low-Reynolds-number validation case before conducting a larger parameter study.
2. Verify the mesh using `checkMesh` and perform systematic mesh-sensitivity studies before interpreting changes in aerodynamic quantities.
3. Perform time-step sensitivity studies for transient simulations and monitor the Courant number.
4. Monitor both residuals and integral quantities such as C_d, C_l, and the Strouhal number. Residual convergence alone is not sufficient for assessing a transient solution.
5. Remove the initial transient before calculating statistically averaged quantities.
6. Monitor mass conservation and wall-resolution quantities such as y^+ for every simulation.
7. Keep the geometry, mesh strategy, boundary conditions, numerical schemes, and post-processing procedure consistent when comparing Reynolds numbers or turbulence models.

8. When comparing numerical schemes or turbulence models, modify only the intended parameter while keeping the remaining setup unchanged.
9. Use sufficiently long simulation times to obtain statistically meaningful mean and fluctuating quantities.
10. Validate both mean and unsteady quantities, including C_d, C_l fluctuations, and Strouhal number, against appropriate benchmark or experimental data.
11. Do not assume that changing the RANS turbulence model alone will reproduce high-Reynolds-number experimental behaviour.
12. For high-Reynolds-number studies, assess the influence of three-dimensional modelling before drawing physical conclusions from two-dimensional simulations.
13. Validate an LES methodology at a selected Reynolds number before extending it to substantially higher Reynolds numbers.
14. Record the complete simulation configuration, including solver, turbulence model, mesh, time step, boundary conditions, convergence criteria, and hardware configuration, to maintain reproducibility.

6.7 Key Contributions

The principal contributions of this work can therefore be summarised as follows:

- A parameterised double O-grid block-structured mesh topology for circular-cylinder flow, implemented entirely using OpenFOAM `blockMesh`.
- An analytically parameterised mesh-generation framework in which the principal geometric and grading quantities are generated from a compact set of independent parameters.
- An automated case-generation and execution workflow based on `cases.csv`, `runCases.sh`, and a protected `baseCase` template.
- Automated convergence monitoring and post-processing tools for evaluating residuals, forces, mass balance, wall resolution, and Reynolds-number trends.
- A systematic numerical investigation covering Reynolds-number variation, temporal discretisation, solver formulation, mesh sensitivity, time-step sensitivity, and parallel execution.
- A comparative investigation of standard k-epsilon and k-omega SST turbulence models and identification of their limitations for the present two-dimensional higher-Reynolds-number configuration.
- A reproducible framework that can be extended to three-dimensional RANS and subsequently LES without replacing the underlying campaign automation structure.

6.8 Closing Remarks

The work has progressed from the development of a parameterised OpenFOAM simulation infrastructure to a systematic investigation of its numerical behaviour and modelling limitations. The results demonstrate that reliable CFD predictions require more than a functioning solver: mesh resolution, time-step size, solver formulation, convergence behaviour, wall resolution, turbulence modelling, and dimensionality must all be considered together.

The lower-Reynolds-number studies established a useful numerical baseline, while the grid and time-step investigations identified practical configurations for further calculations. The turbulence-model studies then demonstrated the limitations of the present two-dimensional RANS approach as the Reynolds number increases.

The next important step is therefore not simply to extend the existing two-dimensional simulations to higher Reynolds numbers, but to increase the physical fidelity of the model through three-dimensional modelling and, where computationally feasible, LES. The existing automated framework provides a reproducible foundation for this continuation and allows the additional simulations to be conducted systematically rather than through manual case-by-case modification.

Ultimately, the objective of the continued study should be to establish a validated numerical methodology that reproduces not only individual aerodynamic coefficients, but also the experimentally observed dependence of drag, vortex shedding, and wake behaviour on Reynolds number.

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A
Appendix

A.1 Case Cleanup Script: `cleanCases.sh`

The `cleanCases.sh` script removes generated CFD cases, logs, and results during repeated simulation campaigns. It automatically determines the project root directory and therefore can be executed from different locations.

Available options:

- `-all`: removes all generated cases.
- `-label <name>`: removes a specific case.
- `-results`: additionally removes the `results/` directory.

The `logs/` directory is cleared during every valid cleanup operation, while the original `baseCase/` remains unaffected.

Examples:

```
./cleanCases.sh --all
./cleanCases.sh --label case_Re500000
./cleanCases.sh --all --results
```

```
#!/bin/bash
#============================================================
# Simple cleanCases.sh
# - Deletes case folders
# - Optional: delete logs + results
# - Works from anywhere
#============================================================

set -e

# ---- Resolve paths ------------------------------------------
SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
ROOT_DIR="$(dirname "$SCRIPT_DIR")"

CASESDIR="$ROOT_DIR/cases"
LOGDIR="$ROOT_DIR/logs"
RESULTS="$ROOT_DIR/results"
```

```
18
19 FILTER_LABEL=""
20 CLEAN_ALL=false
21 CLEAN_RESULTS=false
22
23 # ---- Parse arguments ----------------------------------------
24 while [[ $# -gt 0 ]]; do
25     case "$1" in
26         --all)     CLEAN_ALL=true ;;
27         --label)   FILTER_LABEL="$2"; shift ;;
28         --results) CLEAN_RESULTS=true ;;
29         *) echo "Unknown option: $1"; exit 1 ;;
30     esac
31     shift
32 done
33
34 # ---- Safety checks ------------------------------------------
35 [ ! -d "$CASESDIR" ] && { echo "No cases directory found."; exit 1;
       }
36
37 echo "Project root: $ROOT_DIR"
38 echo "-----------------------------------"
39
40 # ---- Clean logic --------------------------------------------
41 if $CLEAN_ALL; then
42     echo "Removing ALL cases..."
43     rm -rf "$CASESDIR"
44     mkdir -p "$CASESDIR"
45
46 elif [ -n "$FILTER_LABEL" ]; then
47     TARGET="$CASESDIR/$FILTER_LABEL"
48
49     if [ -d "$TARGET" ]; then
50         echo "Removing case: $FILTER_LABEL"
51         rm -rf "$TARGET"
52     else
53         echo "Case not found: $FILTER_LABEL"
54     fi
55
56 else
57     echo "Use --all or --label <name>"
58     exit 1
59 fi
60
61 echo "Cleaning logs..."
62 rm -rf "$LOGDIR"
63 mkdir -p "$LOGDIR"
64
65 if $CLEAN_RESULTS; then
66     echo "Cleaning results..."
```

```
    rm -rf "$RESULTS"
    mkdir -p "$RESULTS"
fi

echo "Done."
```

Listing A.1: Case Cleanup Script: `cleanCases.sh`

A.2 Case Execution Script: `runCases.sh`

The `runCases.sh` script automates the CFD simulation campaign by reading parameters from `cases.csv`, creating cases from `baseCase`, updating the physical and mesh parameters, generating the mesh, and executing the solver in parallel.

Available options:

- `-parallel N`: specifies the number of processor cores.
- `-label <FILTER>`: runs only matching cases.
- `-h, -help`: displays the available options.

For each case, the script updates the velocity, solver settings, time controls, mesh parameters, and time discretisation before executing `blockMesh`, `decomposePar`, the solver, and `reconstructPar`.

Examples:

```
./runCases.sh
./runCases.sh --parallel 8
./runCases.sh --label case_Re500000
./runCases.sh --parallel 8 --label Re500000
```

```
#!/bin/bash
#============================================================
# runCases.sh
# Reads cases.csv, copies baseCase for each entry, updates the
# case parameters (physics + mesh), and runs the solver.
#============================================================

set -euo pipefail

SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
ROOT_DIR="$(dirname "$SCRIPT_DIR")"
BASEDIR="$ROOT_DIR/baseCase"
CASESFILE="$ROOT_DIR/cases.csv"
CASESDIR="$ROOT_DIR/cases"
LOGDIR="$ROOT_DIR/logs"

PARALLEL_JOBS=8
FILTER_LABEL=""

```

```
usage() {
    cat <<EOF
Usage: bash runCases.sh [options]

Options:
  --parallel N      Number of cores for decomposition/solver (
   default: 8)
  --label FILTER    Only run rows whose label contains FILTER
  -h, --help        Show this help message
EOF
}

while [[ $# -gt 0 ]]; do
    case "$1" in
        --parallel)
            PARALLEL_JOBS="$2"
            shift 2
            ;;
        --label)
            FILTER_LABEL="$2"
            shift 2
            ;;
        -h|--help)
            usage
            exit 0
            ;;
        *)
            echo "Unknown option: $1"
            usage
            exit 1
            ;;
    esac
done

mkdir -p "$CASESDIR" "$LOGDIR"

if [ -z "${WM_PROJECT_DIR:-}" ]; then
    source /opt/openfoam10/etc/bashrc 2>/dev/null || true
fi

if [ ! -f "$CASESFILE" ]; then
    echo "ERROR: cases.csv not found at $CASESFILE"
    exit 1
fi

COMPLETED_CASES=()
FAILED_CASES=()

echo "Running CFD campaign..."
echo "-------------------------------"
```

```
69
70 mapfile -t CSV_LINES < <(tail -n +2 "$CASESFILE")
71 echo "Total cases to process: ${#CSV_LINES[@]}"
72
73 for csv_line in "${CSV_LINES[@]}"; do
74     [[ -z "$csv_line" ]] && continue
75
76     IFS=',' read -r Re state solver ddT U endTime dt maxCo D \
77         N_bl E R_prism_mult E_far N_c N_z label <<< "$csv_line"
78
79     for var in Re state solver ddT U endTime dt maxCo D N_bl E \
80         R_prism_mult E_far N_c N_z label; do
81         declare "$var=${!var//$'\r'/}"
82     done
83
84     if [[ -z "$label" || -z "$solver" ]]; then
85         echo "Skipping bad row: $csv_line"
86         continue
87     fi
88
89     if [[ -n "$FILTER_LABEL" && "$label" != *"$FILTER_LABEL"* ]];
    then
90         continue
91     fi
92
93     CASEDIR="$CASESDIR/$label"
94     LOGFILE="$LOGDIR/$label.log"
95
96     if [ -d "$CASEDIR" ]; then
97         echo "Skipping existing case: $label"
98         continue
99     fi
100
101    cp -r "$BASEDIR" "$CASEDIR"
102
103    sed -i "s|INLET_U|$U|g" "$CASEDIR/0/U"
104    sed -i "s|INLET_U|$U|g" "$CASEDIR/system/controlDict"
105    sed -i "s|^application.*|application     $solver;|" \
106        "$CASEDIR/system/controlDict"
107    sed -i "s|^endTime.*|endTime         $endTime;|" \
108        "$CASEDIR/system/controlDict"
109    sed -i "s|^deltaT.*|deltaT          $dt;|" \
110        "$CASEDIR/system/controlDict"
111    sed -i "s|^maxCo.*|maxCo            $maxCo;|" \
112        "$CASEDIR/system/controlDict"
113
114    GEOM="$CASEDIR/system/geometryProperties"
115
116    if [ -f "$GEOM" ]; then
117        update_dict() {
```

```
118             local param=$1
119             local val=$2
120             local file=$3
121             sed -i -E \
122                 "s/^([[:space:]]*${param}[[:space:]]+)[^;]+;/\1${
    val};/g" \
123                 "$file"
124         }
125 
126         update_dict "D" "$D" "$GEOM"
127         update_dict "N_bl" "$N_bl" "$GEOM"
128         update_dict "E" "$E" "$GEOM"
129         update_dict "R_prism_mult" "$R_prism_mult" "$GEOM"
130         update_dict "E_far" "$E_far" "$GEOM"
131         update_dict "N_c" "$N_c" "$GEOM"
132         update_dict "N_z" "$N_z" "$GEOM"
133     fi
134 
135     if [ "$state" = "steady" ]; then
136         sed -i '/ddtSchemes/,/}/c\
137 ddtSchemes
138 {
139     default steadyState;
140 }' "$CASEDIR/system/fvSchemes"
141     else
142         sed -i "/ddtSchemes/,/}/c\
143 ddtSchemes
144 {
145     default $ddT;
146 }" "$CASEDIR/system/fvSchemes"
147     fi
148 
149     (cd "$CASEDIR" && blockMesh >>"$LOGFILE" 2>&1)
150     (cd "$CASEDIR" && decomposePar -force >>"$LOGFILE" 2>&1)
151     (cd "$CASEDIR" && \
152         mpirun -np "$PARALLEL_JOBS" "$solver" -parallel >>"$LOGFILE
    " 2>&1)
153     (cd "$CASEDIR" && reconstructPar >>"$LOGFILE" 2>&1)
154 
155     rm -rf "$CASEDIR"/processor*
156 done
157 
158 echo "Campaign completed."
```

Listing A.2: Case Execution Script: `runCases.sh`

A.3 Post-Processing Script: `postProcess.py`

The `postProcess.py` script automates the analysis of OpenFOAM results for all available cases. It evaluates force coefficients, Strouhal number, y^+, mass-flow balance, and solver residuals, while generating case-wise and global comparison plots.

Main functions:

- Automatic discovery and processing of CFD cases.
- Calculation of C_d, C_l, $C_{l,\mathrm{rms}}$, and St.
- Evaluation of y^+ and mass-flow imbalance.
- Generation of force, FFT, y^+, mass-balance, and residual plots.
- Generation of global comparisons versus Reynolds number.
- Automatic creation of CSV and text summary reports.

Usage:

```
python3 postProcess.py
```

The case-specific analysis is stored in `postProcessing/analysis/`, while global results are stored in `global`$_a$`nalysis/`.

```
#!/usr/bin/env python3

from pathlib import Path
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg')

import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq
import warnings
import re

warnings.filterwarnings("ignore")

SCRIPT_DIR = Path(__file__).resolve().parent
ROOT_DIR = SCRIPT_DIR.parent
CASES_DIR = ROOT_DIR / "cases"

GLOBAL_ANALYSIS_DIR = ROOT_DIR / "global_analysis"
GLOBAL_ANALYSIS_DIR.mkdir(exist_ok=True)

SUMMARY_FILE = GLOBAL_ANALYSIS_DIR / "global_case_summary.csv"

D = 0.1
NU = 1.5e-5

```

```
FIGSIZE = (10, 6)
DPI = 300

# ... rest of your original postProcess.py ...
```

Listing A.3: Post-Processing and Analysis Script: `postProcess.py`

A.4 ParaView File Generation Script: `touch_foam.sh`

The touch_foam.sh script creates a .foam marker file inside each simulation case directory, allowing the corresponding OpenFOAM case to be opened in ParaView.

Functionality:

- Searches for simulation case directories.
- Creates a corresponding .foam file.
- Reports the generated files to the terminal.

Usage:

```
./touch_foam.sh
```

```
#!/bin/bash
# ============================================================
# touch_foam.sh
# Navigate to each case folder and create <folder_name>.foam
# ============================================================

CASES_DIR="../cases"

for d in "$CASES_DIR"/case_Re*; do

    if [ -d "$d" ]; then

        CASE_NAME=$(basename "$d")

        cd "$d" || {
            echo "Failed to enter $d"
            continue
        }

        touch "${CASE_NAME}.foam"
        echo "Created ${CASE_NAME}.foam in $d"

        cd - > /dev/null
    fi

done

echo "All .foam files created for all cases!"
```

Listing A.4: ParaView `.foam` File Generation Script